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ELSEVIER Stochastic Processes and their Applications 115 (2005) 1557–1582
stochastic processes and their applications
www.elsevier.com/locate/spa

The L^2-structures of standard and switching-regime GARCH models

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Received 17 March 2004; received in revised form 21 March 2005; accepted 15 April 2005
Available online 17 May 2005

Abstract

This paper analyzes the probabilistic structure of Markov-switching GARCH(p, q) models, in which the volatility process is driven by a finite state-space Markov chain. We give necessary and sufficient conditions for the existence of moments of any order. We find that the squares and higher order powers of the process have the L^2 structures of ARMA processes, and hence admit ARMA representations. These results are applicable to standard GARCH models and have statistical implications in terms of order identification and parameter estimation.
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MSC: 60J05; 60G10; 60G12; 62M10

Keywords: ARMA representation; GARCH; HMM; Markov-switching models

1. Introduction

In the statistical analysis of returns on financial assets, one is confronted with the task of modeling time series with a very complicated structure. This structure is characterized by a set of empirical features that are common to many financial data,

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doi:10.1016/j.spa.2005.04.005

such as stock prices or indices, and exchange rates. These "stylized facts" are lack of serial correlation, heavy-tailed marginal distributions, volatility clustering (large and small observations tend to appear in clusters), and slow decrease of the sample autocorrelations of the squares (or absolute values) of the data. These properties make standard time series models, such as ARMA processes, inappropriate for such series.

The most popular class of volatility models is the autoregressive conditional heteroskedasticity (ARCH) specification introduced by Engle [11] and generalized (GARCH) by Bollerslev [5]. There is a still growing literature on GARCH and related models (see e.g. [2,3,16,21,23] for recent references; see [27] for a comprehensive review on GARCH and stochastic volatility processes). These models have the form $\varepsilon_t = \sqrt{h_t}\eta_t$, where (h_t) is a positive stochastic process, called volatility, and (η_t) is an iid centered sequence, with h_t independent of η_t. Under moment conditions (ε_t) is an uncorrelated sequence. In the standard model, h_t is specified as a linear function of past values of ε_t^2 and h_t. GARCH models gained popularity among practitioners, and in the econometric literature, because they often give a sensible fit to many types of financial data of moderate sample size. However, empirical work has shown that these models have failed to fit the data very well over a long period of time: parameters change, requiring the practitioners to reestimate them.

This observation leads to Markov-switching (MS) GARCH models: GARCH models in which the volatility parameters are assumed to depend on a Markov chain. The MS GARCH(p, q) model we consider is defined by

$$\begin{aligned} \varepsilon_t &= \sqrt{h_t}\eta_t, \\ h_t &= \omega(\Delta_t) + \sum_{i=1}^{q} \alpha_i(\Delta_t)\varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j(\Delta_t)h_{t-j}, \end{aligned} \tag{1}$$

where the parameters are functions of a finite state ergodic Markov chain, denoted by (Δ_t), which is independent of the centered iid sequence (η_t). Precise definitions and assumptions are given in Section 2. MS GARCH models generate series with a much more flexible dependence structure than in standard GARCH specifications. Since the seminal paper by Hamilton [18], the use of Markov-Switching Models has become increasingly popular in econometrics. GARCH models of the type (1), with Markov-switching coefficients, have already been discussed by, for example, Hamilton and Susmel [19], Cai [9], Gray [17], Dueker [10], Francq et al. [12]. A particular case of interest is when (Δ_t) is an independent process (see [30]).

The purpose of this paper is to investigate the L^2 structure of the MS GARCH processes and their even powers. This is a substantial mathematical problem due to the presence of a latent process in the volatility equation. We find that, under existence conditions, the L^2 structure of (ε_t^{2m}), for any nonnegative integer m, is that of a finite ARMA model. This is in particular the case for standard GARCH models. To our knowledge this result was known only for standard GARCH and $m = 1$. We are able to characterize the orders of this ARMA in terms of the MS GARCH

parameters. The method for deriving these ARMA representations is novel and may find applications in other contexts.

The L^2-structure results presented here relate to a literature dealing with MS ARMA models. Francq and Zakoïan [13] establish necessary and sufficient conditions for the existence of strict and second-order stationary solutions to MS ARMA models. Karlsen [22], Poskitt and Chung [29], and Zhang and Stine [31], obtain ARMA representations for autoregressive processes governed by a Markov chain. However, these results do not apply to our framework for the following reason. Although a squared standard GARCH process follows an ARMA model, a squared MS GARCH process does not in general follow a MS ARMA model.

The results in the present paper can be applied to identify the orders and to estimate the parameters of model (1). Given the untractability of the likelihood in this framework, it is extremely important to develop alternative methods. In Francq and Zakoïan [14], motivated by the structural results of the present paper, we investigate an estimation method. The idea is to select the MS model that better matches the second-order behavior of the observed series and its powers.

The article is organized as follows. Section 2 introduces the main assumptions on the model and derives necessary and sufficient conditions for the existence of moments. Section 3 deals with the autocovariance structure of (ε_t^2). The existence of an ARMA representation is proved and its orders are explicitly given. Section 4 extends these results to the powers of (ε_t^2). We establish the existence of an ARMA representation for (ε_t^{2m}) and characterize the orders and parameters of this representation.

2. Assumptions and moment conditions

Throughout this paper we make the following standard assumption.

Assumption A. The processes (η_t) and (Δ_t) are independent; (η_t) is a sequence of iid random variables with zero mean and unit variance; (Δ_t) is a homogeneous, stationary, irreducible, and aperiodic Markov chain with finite state-space $\mathscr{E} = \{1, 2, \ldots, d\}$; the coefficients satisfy the constraints $\omega(k) > 0$, $\alpha_i(k) \geqslant 0$, and $\beta_j(k) \geqslant 0$ for $k \in \mathscr{E}$, $i \in \{1, \ldots, q\}$, $j \in \{1, \ldots, p\}$.

The stationary probabilities of (Δ_t) are denoted by $\pi(k) := P(\Delta_1 = k)$ and are positive under Assumption A, the transition probabilities are denoted by $p(k, \ell) := P(\Delta_t = \ell | \Delta_{t-1} = k)$, and the i-step transition probabilities by $p^{(i)}(k, \ell) = P(\Delta_t = \ell | \Delta_{t-i} = k)$ for $k, \ell \in \mathscr{E}$ and $i \geqslant 1$.

Consider the matrices

$$\mathbb{P}_f^{(i)} = \begin{pmatrix} p^{(i)}(1,1)f(1) & \cdots & p^{(i)}(d,1)f(1) \\ \vdots & & \vdots \\ p^{(i)}(1,d)f(d) & \cdots & p^{(i)}(d,d)f(d) \end{pmatrix} \quad \text{and} \quad \Pi_f = \begin{pmatrix} \pi(1)f(1) \\ \vdots \\ \pi(d)f(d) \end{pmatrix} \tag{2}$$

for any function $f: \mathscr{E} \to \mathscr{M}_{n\times n'}(\mathbb{R})$, where $\mathscr{M}_{n\times n'}(\mathbb{R})$ denotes the space of real $n \times n'$ matrices, and any positive integers i, n and n'. When $i = 1$ we simply use $\mathbb{P}_f$ to signify $\mathbb{P}_f^{(1)}$, and when $f \equiv 1$ we use $\mathbb{P} := \mathbb{P}_1 = (p(\ell, k))$ to denote the transpose of the transition matrix. Note that when $n = n'$, $\Pi_f = \mathbb{P}_f \Pi_{I_n}$, where I_n is the $n \times n$ identity matrix. Note also that

$$\text{when } n = n' = 1, \quad \det\{\mathbb{P}_f^{(i)}\} = \{\det \mathbb{P}\}^i \prod_{k=1}^{d} f(k) \tag{3}$$

and that, still when $n = n' = 1$

$$\text{when } \prod_{k=1}^{d} f(k) \neq 0, \quad \operatorname{rank}\{\mathbb{P}_f^{(i)}\} = \operatorname{rank}(\mathbb{P}^i) \quad \text{for any } i \geqslant 1. \tag{4}$$

We now derive necessary and sufficient conditions for the existence of a strictly stationary solution to model (1) with a finite $2m$th order moment, for $m = 1, 2, \ldots$. Along the lines of the work done by Bougerol and Picard [7] for standard GARCH models, we consider the model

$$\underline{z}_t = \underline{b}_t + C_t \underline{z}_{t-1}, \tag{5}$$

where $\underline{b}_t = (\omega(\Delta_t)\eta_t^2, 0, \ldots, 0, \omega(\Delta_t), 0, \ldots, 0)' \in \mathbb{R}^{p+q}$, $\underline{z}_t \in \mathbb{R}^{p+q}$, and

$$C_t = \begin{pmatrix} \boldsymbol{\alpha}_{1:q-1}(\Delta_t)\eta_t^2 & \alpha_q(\Delta_t)\eta_t^2 & \boldsymbol{\beta}_{1:p-1}(\Delta_t)\eta_t^2 & \beta_p(\Delta_t)\eta_t^2 \\ I_{q-1} & 0 & 0 & 0 \\ \boldsymbol{\alpha}_{1:q-1}(\Delta_t) & \alpha_q(\Delta_t) & \boldsymbol{\beta}_{1:p-1}(\Delta_t) & \beta_p(\Delta_t) \\ 0 & 0 & I_{p-1} & 0 \end{pmatrix}$$

where $\boldsymbol{\alpha}_{1:q-1}(\Delta_t) = (\alpha_1(\Delta_t), \ldots, \alpha_{q-1}(\Delta_t))$, $\boldsymbol{\beta}_{1:p-1}(\Delta_t) = (\beta_1(\Delta_t), \ldots, \beta_{p-1}(\Delta_t))$. When $p = 0$ or $q = 0$ obvious modifications have to be done in the definitions of $\underline{b}_t$ and C_t. There is a one-to-one correspondence between the solutions (ε_t) of (1) and the positive solutions $(\underline{z}_t)$ of (5). If (ε_t) is solution to (1), take $\underline{z}_t = (\varepsilon_t^2, \ldots, \varepsilon_{t-q+1}^2, h_t, \ldots, h_{t-p+1})'$. Conversely, if $(\underline{z}_t)$ is a solution of (5) with positive components, take ε_t equal to the square root of the first component multiplied by the sign of η_t.

Before stating the main result of this section we need some further notation. Denote by $\rho(A)$ the spectral radius of any square matrix A. Let $\otimes$ denote the Kronecker product and, for any matrix A, let $A^{\otimes m} = A \otimes \cdots \otimes A$. A square positive matrix A is called irreducible if for all i, j there exists a positive integer n such that $A^{(n)}(i, j) > 0$, where $A^{(n)}(i, j)$ denotes the generic element of A^n. A process (X_t) is called nonanticipative if, for all t, X_t is measurable with respect to the σ-field generated by $\{\Delta_u, \eta_u;\ u \leqslant t\}$. In the statement of the results we will index the expectations by the processes involved, when necessary to avoid ambiguity.

We begin by stating a preliminary result which can be easily proved.

Lemma 1. *Let $f:\ \mathscr{E}\to\mathscr{M}_{n\times n}(\mathbb{R})$ and $g:\ \mathscr{E}\to\mathscr{M}_{n\times n'}(\mathbb{R})$. Then, for $k>0$, and $h>k$,*

$$E\{f(\Delta_t)f(\Delta_{t-1})\dots f(\Delta_{t-k+1})g(\Delta_{t-k})\mid \Delta_{t-h}\}=\mathbb{I}\mathbb{P}_f^k\mathbb{P}_g^{(h-k)}(e_{\Delta_{t-h}}\otimes I_{n'}),$$

$$E\{f(\Delta_t)f(\Delta_{t-1})\dots f(\Delta_{t-k+1})g(\Delta_{t-k})\}=\mathbb{I}\mathbb{P}_f^k\Pi_g,$$

where $\mathbb{I}=(I_n,\dots,I_n)$ is a $n\times nd$ matrix and (e_i) denotes the canonical basis of $\mathbb{R}^d$.

We are now ready to state the following theorem.

Theorem 1. *Suppose that $E(\eta_t^{2m})<\infty$ and*

$$\rho\{\mathbb{P}_{C^{(m)}}\}<1, \tag{6}$$

where $C^{(m)}(\cdot)=E_\eta(C_t^{\otimes m}|\Delta_t=\cdot)$ and m is a strictly positive integer. Then (1) *has a unique nonanticipative and strictly stationary solution* (ε_t). *This solution is also ergodic and satisfies $E(\varepsilon_t^{2m})<\infty$.*

Conversely, if $\mathbb{P}_{C^{(m)}}$ is irreducible and $\rho\{\mathbb{P}_{C^{(m)}}\}\geqslant 1$, then there is no strictly stationary solution (ε_t) *to model* (1) *such that $E(\varepsilon_t^{2m})<\infty$.*

Proof. First note that there exists a solution (ε_t) of (1) belonging to L^{2m} if and only if there exists a solution $\underline{z}_t$ of (5) belonging to L^m. The proof is divided in three steps:

(i) We show that (6) is a sufficient condition for the existence of a solution of (5) belonging to L^m. The vector

$$\underline{z}_t^*=\underline{b}_t+\sum_{k=1}^{\infty}C_tC_{t-1}\cdots C_{t-k+1}\underline{b}_{t-k} \tag{7}$$

is well-defined (its components belong to $\mathbb{R}^+\cup\{+\infty\}$ with probability one, because the components of b_t and $C_tC_{t-1}\cdots C_{t-k+1}\underline{b}_{t-k}$ are nonnegative). For the series in (7) to belong to L^m, it suffices that

$$\underline{z}_{t,k}:=C_tC_{t-1}\cdots C_{t-k+1}\underline{b}_{t-k}$$

(with $\underline{z}_{t,0}=\underline{b}_t$) converges to zero in L^m (endowed with any matrix norm) at an exponential rate, as k goes to infinity. For any vector X such that AX is well defined, we have $(AX)^{\otimes m}=A^{\otimes m}X^{\otimes m}$. It readily follows that

$$\underline{z}_{t,k}^{\otimes m}=C_t^{\otimes m}C_{t-1}^{\otimes m}\cdots C_{t-k+1}^{\otimes m}\underline{b}_{t-k}^{\otimes m}. \tag{8}$$

Thus we may write, in view of the independence of the matrices and vector $C_t,\dots,C_{t-k+1}$, $\underline{b}_{t-k}$ conditional on Δ_t,

$$\begin{aligned}E(\underline{z}_{t,k}^{\otimes m})&=E\{E(C_t^{\otimes m}C_{t-1}^{\otimes m}\cdots C_{t-k+1}^{\otimes m}\underline{b}_{t-k}^{\otimes m}|\Delta_t,\dots,\Delta_{t-k})\}\\&=E\{C^{(m)}(\Delta_t)\dots C^{(m)}(\Delta_{t-k+1})\underline{b}^{(m)}(\Delta_{t-k})\}\end{aligned}$$

where $C^{(m)}(\Delta_t)=E(C_t^{\otimes m}|\Delta_t)$ and $\underline{b}^{(m)}(\Delta_t)=E(\underline{b}_t^{\otimes m}|\Delta_t)$. Applying Lemma 1 with $n=(p+q)^m$, we obtain

$$E(\underline{z}_{t,k}^{\otimes m})=\mathbb{I}\{\mathbb{P}_{C^{(m)}}\}^k\Pi_{\underline{b}^{(m)}}, \tag{9}$$

Consider the matrix norm defined by $\|A\|=\sum_{i,j}|A(i,j)|$, where $A(i,j)$ denotes the generic element of some matrix A. Using the elementary equalities

$\|A\|\|B\| = \|A \otimes B\| = \|B \otimes A\|$, it holds that $\|\underline{z}_{t,k}\|^m = \|\underline{z}_{t,k}^{\otimes m}\|$. Hence $E\|\underline{z}_{t,k}\|^m = E\|\underline{z}_{t,k}^{\otimes m}\| = \|E(\underline{z}_{t,k}^{\otimes m})\|$ because $E\|X\| = \|E(X)\|$ for any vector with positive components. In view of (9), we thus have

$$E\|\underline{z}_{t,k}\|^m = \|\mathbb{I}\{\mathbb{P}_{C^{(m)}}\}^k \Pi_{\underline{b}^{(m)}}\|. \tag{10}$$

The matrix norm being multiplicative, we have

$$\|\underline{z}_{t,k}\|_{L^m} = \|\mathbb{I}\{\mathbb{P}_{C^{(m)}}\}^k \Pi_{\underline{b}^{(m)}}\|^{1/m} \leqslant \|\mathbb{I}\|^{1/m} \|\{\mathbb{P}_{C^{(m)}}\}^k\|^{1/m} \|\Pi_{\underline{b}^{(m)}}\|^{1/m}.$$

If (6) holds, it follows from the Jordan decomposition that $\|\mathbb{P}_{C^{(m)}}^k\|$ converges to zero at an exponential rate as k goes to infinity. Hence, $\sum_{k=0}^K \underline{z}_{t,k}$ converges to $\underline{z}_t^*$ defined by (7), both in L^m and almost surely, for fixed t, as $K \to \infty$. For each K, $(\sum_{k=0}^K \underline{z}_{t,k})$ is a measurable function of $(\Delta_t, \eta_t, \ldots, \Delta_{t-K}, \eta_{t-K})$. Hence the limit $(\underline{z}_t^*)$ is a measurable function of the strictly stationary process (Δ_t, η_t). From Billingsley [4, Theorem 36.4] it follows that $(\underline{z}_t^*)$ is strictly stationary. It is clear that $\underline{z}_t^* = \underline{b}_t + C_t \underline{z}_{t-1}^*$.Therefore, (5) admits a nonanticipative strictly stationary solution belonging to L^m. Moreover, this solution is ergodic because (Δ_t, η_t) is an ergodic stationary process.

(ii) Now assume that another strictly stationary solution of (1) exists, or equivalently that there is another strictly stationary solution $(\tilde{\underline{z}}_t)$ of (5). Then for any $n \geqslant 0$

$$\|\tilde{\underline{z}}_t - \underline{z}_t^*\| = \|C_t(\tilde{\underline{z}}_{t-1} - \underline{z}_{t-1}^*)\| = \cdots \leqslant \|C_t \ldots C_{t-n}\| \, \|\tilde{\underline{z}}_{t-n-1} - \underline{z}_{t-n-1}^*\|.$$

By assumption $\mathbb{P}(\|\tilde{\underline{z}}_t - \underline{z}_t^*\| \neq 0) > 0$. But we know that $\|C_t \ldots C_{t-n}\| \to 0$ a.s. when $n \to \infty$ since the sum in (7) converges. It follows that $\mathbb{P}(\lim_{n\to\infty} \|\tilde{\underline{z}}_{t-n-1} - \underline{z}_{t-n-1}^*\| = \infty) > 0$, which entails that either $\limsup_{n\to\infty} \|\tilde{\underline{z}}_{t-n-1}\| = \infty$ or $\limsup_{n\to\infty} \|\underline{z}_{t-n-1}^*\| = \infty$ with a positive probability. This is absurd since the sequences $(\tilde{\underline{z}}_t)_t$ and $(\underline{z}_t^*)_t$ are stationary.The uniqueness of $(\underline{z}_t^*)_t$ is thus established.

(iii) We now prove the necessary part. For two matrices A and B of same size, we use the notation $A \succcurlyeq B$ (resp. $A \succ B$) to indicate that the (real) components of $A - B$ are nonnegative (resp. strictly positive). In view of (5), for any k

$$\underline{z}_t = \underline{z}_{t,0} + \cdots + \underline{z}_{t,k} + C_t \ldots C_{t-k} \underline{z}_{t-k-1} \succcurlyeq \sum_{\ell=0}^{k} \underline{z}_{t,\ell}$$

because the matrices C_t and the vectors $\underline{z}_t$ have nonnegative components. Hence $\underline{z}_t \succcurlyeq \sum_{\ell=0}^{\infty} \underline{z}_{t,\ell}$ and

$$E(\underline{z}_t^{\otimes m}) \succcurlyeq E\left(\sum_{\ell=0}^{\infty} \underline{z}_{t,\ell}\right)^{\otimes m} \succcurlyeq \sum_{\ell=0}^{\infty} E(\underline{z}_{t,\ell}^{\otimes m}) = \sum_{\ell=0}^{\infty} \mathbb{I}\{\mathbb{P}_{C^{(m)}}\}^\ell \Pi_{\underline{b}^{(m)}} \tag{11}$$

using (9). For any irreducible $n \times n$ matrix $A \succcurlyeq 0$, we have $(I + A)^{n-1} \succ 0$ (see [25, p. 533]). In addition, the elementary relations $\|\mathbb{I}A\| = \|A\|$, $\|A\| \geqslant \rho(A)$, $\rho(A^n) = \{\rho(A)\}^n$, $n!(I + \cdots + A^n) \succcurlyeq (I + A)^n$ and $\|A + B\| = \|A\| + \|B\|$ hold, for any matrices $A \succcurlyeq 0$ and $B \succcurlyeq 0$ of appropriate dimensions. Moreover, $\|\sum_{\ell=0}^{\infty} A_\ell\| = \sum_{\ell=0}^{\infty} \|A_\ell\|$ for any sequence of matrices $A_\ell \succcurlyeq 0$ of the same dimension, by continuity of the norm.

Therefore, by (11)

$$\begin{aligned}\|E(\underline{z}_t^{\otimes m})\| &\geqslant \left\| \sum_{\ell=0}^{\infty} \mathbb{I}\{\mathbb{P}_{C^{(m)}}\}^{\ell} \Pi_{\underline{b}^{(m)}} \right\| \\ &= \sum_{\ell=0}^{\infty} \|\mathbb{I}\{\mathbb{P}_{C^{(m)}}\}^{\ell} \Pi_{\underline{b}^{(m)}}\| \\ &= \sum_{\ell=0}^{\infty} \|\{\mathbb{P}_{C^{(m)}}\}^{\ell d(p+q)^m} \{I + \mathbb{P}_{C^{(m)}} + \cdots + \mathbb{P}_{C^{(m)}}^{d(p+q)^m-1}\} \Pi_{\underline{b}^{(m)}}\| \\ &\geqslant \frac{1}{\{d(p+q)^m - 1\}!} \sum_{\ell=0}^{\infty} \|\{\mathbb{P}_{C^{(m)}}\}^{\ell d(p+q)^m} \{I + \mathbb{P}_{C^{(m)}}\}^{d(p+q)^m-1} \Pi_{\underline{b}^{(m)}}\| \\ &\geqslant \frac{c}{\{d(p+q)^m - 1\}!} \sum_{\ell=0}^{\infty} \{\rho\{\mathbb{P}_{C^{(m)}}\}\}^{\ell d(p+q)^m}, \end{aligned} \tag{12}$$

where $c>0$ is the smallest element of $\{I + \mathbb{P}_{C^{(m)}}\}^{d(p+q)^m-1} \Pi_{\underline{b}^{(m)}}$. Hence, because (ε_t) belongs to L^{2m}, the left-hand side of (12) is finite. Therefore, (6) must hold. □

Remark 1. Instead of using (5), one could consider a recursive system with smaller dimension, $\underline{h}_t = \underline{\omega}_t + A_t \underline{h}_{t-1}$, involving only $\underline{h}_t = (h_t, \ldots, h_{t-r+1})$ with $r = \max\{p, q\}$. The latter system was used by Francq et al. [12] to obtain the strict stationarity condition. However, in this representation the matrices A_t are dependent conditional on Δ_t, unless if $p = q = 1$ (A_t is in fact a function of $(\Delta_t, \eta_{t-1}, \ldots, \eta_{t-r})$). For the moment existence, the fact that the matrices C_t are independent conditional on Δ_t is crucial (see the argument after (8)). For this reason we used the representation on $\underline{z}_t$ (see however the proof of Corollary 2 below for the case $p = q = 1$).

In some special cases the condition for $E(\varepsilon_t^{2m}) < \infty$ can be simplified.

Corollary 1. *When either* (i) *the functions* $\alpha_i(\cdot)$ *and* $\beta_j(\cdot)$ *are constant for all* i *and* j (*but not necessarily the intercept* $\omega(\cdot)$), *or* (ii) (Δ_t) *is an independent process*, (6) *can be replaced by*

$$\rho\{E_{\eta,\Delta}(C_t^{\otimes m})\} < 1. \tag{13}$$

Remark 2. In case (i), considered by Hamilton and Susmel [19] and Cai [9], $C_t^{\otimes m}$ depends on η_t only, whereas it depends on η_t and Δ_t in the independent-switching model. Hence, though this condition is apparently the same for the two cases, it does not imply the same restriction on the parameters. Considering the sub-case where the intercept is constant, this corollary applies to the standard GARCH(p, q). In this case, Ling and McAleer [26] have shown that (13) is also necessary.

Proof. Case (i) is easily proved, noting that C_t does not depend on Δ_t. Consider case (ii). Using (8) and the independence of the matrices $C_t^{\otimes m}$, we obtain

$$E\|\underline{z}_{t,k}\|^m = \|E\underline{z}_{t,k}^{\otimes m}\| = \|\{E(C_t^{\otimes m})\}^k E\{\underline{b}^{(m)}(\Delta_t)\}\|.$$

Along the lines of proof of (i) in Theorem 1, we conclude that (13) is a sufficient condition for the existence of a strictly stationary solution with a finite $2m$th order moment. □

Corollary 2. *When $p = q = 1$, a necessary and sufficient condition for the existence of $E\varepsilon_t^{2m}$ is*

$$\rho[\mathbb{P}_{E_\eta\{\alpha_1(\cdot)\eta_t^2+\beta_1(\cdot)\}^m}] < 1. \tag{14}$$

Moreover, if (Δ_t) is independent this condition takes the form

$$E_{\eta,\Delta}\{\alpha_1(\Delta_t)\eta_t^2 + \beta_1(\Delta_t)\}^m < 1. \tag{15}$$

Remark 3. When $\beta_1(\cdot) = 0$, (15) reduces to $E\{\alpha_1(\Delta_t)\}^m E(\eta_t^{2m}) < 1$. This condition was obtained by Wong and Li [30, Theorem 3] for $m = 2$.

Proof. Instead of applying Theorem 1, we give a self-contained proof of Corollary 2. When $p = q = 1$, the vector equation (5) is equivalent to the one-dimensional equation: $h_t = \underline{b}_t + c_t h_{t-1}$, with $\underline{b}_t = \omega(\Delta_t)$ and $c_t = \alpha_1(\Delta_t)\eta_{t-1}^2 + \beta_1(\Delta_t)$. Therefore, the theorem applies with $\mathbb{P}_{C^{(m)}}$ replaced by $\mathbb{P}_{c^{(m)}} := \mathbb{P}_{E\{\alpha_1(\cdot)\eta_t^2+\beta_1(\cdot)\}^m}$. In the proof of the necessary part, the analogous of (11) is given by

$$Eh_t^m \geqslant \sum_{\ell=0}^{\infty} \mathbb{I}\{\mathbb{P}_{c^{(m)}}\}^\ell \Pi_{\underline{b}^{(m)}}.$$

Because $\Pi_{\underline{b}^{(m)}} = \Pi_{\{\omega(\cdot)\}^m} \succ 0$, the arguments given to show (12) lead to the conclusion that, in the MS GARCH(1,1) case, (6) is also necessary.

When (Δ_t) is independent, for any non null function $f : \mathscr{E} \to \mathbb{R}$, the rank of $\mathbb{P}_f$ is 1. Thus, $\rho\{\mathbb{P}_f\} = \mathrm{Tr}\{\mathbb{P}_f\} = Ef(\Delta_t)$, where $\mathrm{Tr}(\cdot)$ denotes the trace of a square matrix. The last part of the proof follows. □

Remark 4. In the case $p = q = 1$, (14) is equivalent to (6). Indeed the (i,j)th block of the matrix $\mathbb{P}_{C^{(m)}}$ is $p(j,i)E(\eta_1^2)^{\otimes m}(\alpha_1(i),\beta_1(i))^{\otimes m}$. It follows that the blocks of $\mathbb{P}_{C^{(m)}}$ are diagonalized by the same eigenvectors matrix, so that we can write $\mathbb{P}_{C^{(m)}} = M\mathbb{P}_{D^{(m)}}M^{-1}$, for some nonsingular matrix M, and where for any k, $D^{(m)}(k)$ is diagonal of rank 1, with non-zero term equal to $E\{\alpha_1(k)\eta_t^2 + \beta_1(k)\}^m$. The equivalence between the conditions (14) and (6) easily follows. This gives an alternative proof for the sufficient part of Corollary 2.

We now turn to the autocovariance structure of (ε_t^{2m}). We will first establish the existence of an ARMA representation for (ε_t^2), which can be done in a more straightforward way than for higher powers. The representation will also be more explicit in that particular case.

3. Autocovariance structure of (ε_t^2)

That (ε_t^2) has the autocorrelation structure of an ARMA$(\max\{p,q\},p)$ is a well-known feature of standard GARCH(p,q) processes. This was first proved by Milhøj

[28] in the ARCH case, and extended to the GARCH by Bollerslev [5,6]. In this section, we show that this feature still holds when the GARCH parameters are allowed to switch in regime.

An approach for deriving ARMA representations of MS models is to find a state-space representation in which the state equation is a Vector AR(1) (see [24]). This method was used for models of the form $X_t = \mu(\Delta_t) + \eta_t$ by Poskitt and Chung [29]. Unfortunately, this method does not seem to apply to our general model (1).

We start by describing an alternative approach for characterizing the existence of ARMA models.

3.1. Preliminary result

A useful result for deriving ARMA representations is the following. Recall that a second-order stationary process (X_t) is called purely nondeterministic if $\bigcap_{-\infty}^{\infty} \mathscr{H}_n(X) = \{0\}$, where $\mathscr{H}_n(X)$ is the Hilbert space spanned by the X_t, $t \leqslant n$ (see [20, p. 137]).

Lemma 2. *Let (X_t) be a real-valued second-order stationary process, with autocovariance function $\gamma(\cdot)$. Suppose that for some $\ell_0 \in \{0, 1, \ldots\}$*

$$\begin{aligned} &\gamma(\ell) = \delta' W(\ell), \quad \forall \ell, \\ &W(\ell) = A W(\ell - 1), \quad \forall \ell > \ell_0, \end{aligned} \tag{16}$$

where the $W(\ell)$ and δ are column-vectors of size n and A is a $n \times n$ matrix. Let $\mathscr{P}$ denote any real polynomial of degree n such that $\delta' \mathscr{P}(A) = 0$. Write

$$\mathscr{P}(x) = x^{n_0} \mathscr{P}_1(x) \mathscr{P}_2(x), \quad n_0 \geqslant 0,$$

where $\mathscr{P}_1$ is a polynomial of degree n_1 whose roots are of modulus 1, and $\mathscr{P}_2$ has no root of modulus 1 or equal to 0. Then (X_t) admits an ARMA$(n - n_0, n + \ell_0 - 1)$ *representation, with AR polynomial $x^n \mathscr{P}(x^{-1})$.*

Moreover if (X_t) is purely nondeterministic, it admits an ARMA$(n - n_0 - n_1, n + \ell_0 - 1 - n_1)$ *representation, with AR polynomial $x^{n - n_0 - n_1} \mathscr{P}_2(x^{-1})$.*

Proof. First note that the existence of $\mathscr{P}$ is guaranteed by the fact that the characteristic polynomial of A cancels when applied to A, by the Cayley–Hamilton theorem. However, we do not assume that $\mathscr{P}$ is the characteristic polynomial of A. It is not restrictive to assume that $\mathscr{P}(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_{n_0} x^{n_0}$ with $a_n = 1$. Since $W(\ell) = A^{\ell - \ell_0} W(\ell_0)$, $\ell \geqslant \ell_0$, we find, for all $\ell > n + \ell_0 - 1$,

$$\sum_{i=0}^{n-n_0} a_{n-i} \gamma(\ell - i) = \delta' \sum_{i=0}^{n-n_0} a_{n-i} A^{\ell - \ell_0 - i} W(\ell_0) = \delta' \mathscr{P}(A) A^{\ell - \ell_0 - n} W(\ell_0) = 0.$$

In view of the standard result characterizing the existence of an ARMA representation through the autocovariance function (see e.g. [8, Proposition 3.2.1 and Remark, p. 90]), we conclude that (X_t) admits an ARMA$(n - n_0, n + \ell_0 - 1)$

representation, with AR polynomial

$$\sum_{i=0}^{n-n_0} a_{n-i}x^i = x^n\mathscr{P}(x^{-1}) := x^{n-n_0-n_1}P_1(x)\mathscr{P}_2(x^{-1}),$$

where $P_1(x) = x^{n_1}\mathscr{P}_1(x^{-1})$ is a polynomial with all its roots on the unit circle. By Azencott and Dacunha-Castelle [1, Theorem 2.3], when (X_t) is purely nondeterministic we have that

(i) the MA polynomial Q can be factorized as $Q(x) = Q_1(x)P_1(x)$, and
(ii) the common roots of modulus 1 in the AR and MA polynomials can be cancelled.

It follows that (X_t) admits an ARMA representation with AR polynomial $x^{n-n_0-n_1}\mathscr{P}_2(x^{-1})$, and MA polynomial $Q_1(x)$. The conclusion follows. □

Remark 5. It is important to note that the ARMA representation of the theorem is not necessarily the *canonical* ARMA representation, i.e. the AR and MA polynomials may have common roots, or roots inside the unit disk. The canonical representation can be straightforwardly deduced by cancelling the common roots and by inverting the roots of modulus smaller than 1 (see e.g. [8, Remark 1(a), p. 86 and Proposition 3.5.1]).

Remark 6. The reciprocal of this result is trivial: if (X_t) is a second-order stationary ARMA(p,q) process, then a representation of the form (16) exists, with $W(\ell) = (\gamma(\ell),\dots,\gamma(\ell-p+1))'$, $\delta = (1,0,\dots,0)$, and $\ell_0 = q$. To apply the lemma, the difficulty is to find $W(\ell)$, δ and A for which a relation of the form (16) can be exhibited. This will be illustrated below.

3.2. ARMA representation for (ε_t^2)

In this section we will show that the square of any second-order stationary solution of model (1) admits an ARMA representation. Let $r = \max(p,q)$, for all $i \leqslant r$

$$\xi_i(\cdot) = \alpha_i(\cdot) + \beta_i(\cdot) \tag{17}$$

with by convention $\alpha_i(\cdot) = 0$ (resp. $\beta_i(\cdot) = 0$) when $i > q$ (resp. $i > p$). Recall that $\omega(\cdot)$ plays the role of an intercept in model (1). Let the $d(r+1)\times d(r+1)$ matrix

$$A = \begin{pmatrix} \mathbb{P}^{(1)}_{\xi_1} & \cdots & \cdots & \mathbb{P}^{(r)}_{\xi_r} & \mathbb{P}_\omega \\ I_d & 0 & \cdots & \cdots & 0 \\ \vdots & \ddots & & & \vdots \\ 0 & \cdots & I_d & 0 & 0 \\ 0 & \cdots & 0 & 0 & \mathbb{P} \end{pmatrix}. \tag{18}$$

Denote by L the lag operator ($L^i X_t = X_{t-i}$ and $L^i f(t) = f(t-i)$ for any integers i and t, any process (X_t) and any function f defined on the integers set). Denote by $\gamma_X(.)$ the autocovariance function of any stationary process $X = (X_t)$. Let $n_0(M)$ denote the algebraic multiplicity of the zero eigenvalue of any square matrix M. Let $n_1(M)$ be the sum of the algebraic multiplicities of the eigenvalues of M of modulus 1.

Theorem 2. *Suppose that* (6) *holds with* $m = 1, 2$, *and let* (ε_t) *be the nonanticipative fourth-order stationary solution of model* (1).

Then (ε_t^2) *is an ARMA process, with AR order* $N := d(r+1) - n_0(A) - n_1(A)$, *where* $r = \max(p, q) \geqslant 1$, *and MA order* N *when* $p \geqslant q$, *MA order* $N-1$ *when* $p < q$.

Remark 7. As will be seen in the proof, the AR polynomial can be deduced from the characteristic polynomial of A. First note that $n_1(A) \geqslant 1$, because 1 is an eigenvalue of $\mathbb{P}$, and that

$$\det(A) = (-1)^{d(r-1)} \det(\mathbb{P}) \det(\mathbb{P}^{(r)}_{\xi_r}) = (-1)^{d(r-1)} \{\det(\mathbb{P})\}^{r+1} \prod_{k=1}^{d} \xi_r(k)$$

in view of (3). Moreover it can be seen, using (4), that $\operatorname{rank}(A) = (r-1)d + \operatorname{rank}\{\mathbb{P}^{(r)}_{\xi_r}\} + \operatorname{rank}(\mathbb{P})$, i.e.

$$n_0(A) = n_0\{\mathbb{P}^{(r)}_{\xi_r}\} + n_0(\mathbb{P}). \tag{19}$$

The characteristic polynomial of A, defined by $\det(xI - A)$, is therefore of the form

$$\mathscr{P}_A(x) = x^{n_0(A)} \prod_{i=1}^{n_1(A)} (x - \lambda_i) \mathscr{P}_2(x), \quad \mathscr{P}_2(0) \neq 0, \tag{20}$$

where the λ_i are the eigenvalues of modulus 1 of A, and $\mathscr{P}_2$ is a polynomial. We will see in the proof below that the roots of $\mathscr{P}_2$ are inside the unit circle. It will be also shown that the AR polynomial of the ARMA model is $L^{d(r+1) - n_0\{\mathbb{P}^{(r)}_{\xi_r}\} - n_0(\mathbb{P}) - n_1(A)} \mathscr{P}_2(L^{-1})$.

Remark 8. The orders given by the theorem are only maximal ones. However, the only possibility to reduce the AR order occurs when there is a common root of modulus greater than 1 in the AR and MA parts. In any case, the order of the AR part is therefore, after reduction, greater than (or equal to) that of the MA part. This deserves being stressed since it generalizes a standard GARCH property. In other words, if the AR and MA parts have no common root, then the AR order given by the theorem is minimal.

To establish the theorem, the following basic lemma will be used. We give the proof for the sake of completeness.

Lemma 3. *Let, for $i \geqslant 1$, X_{t-i} be an integrable variable belonging to the σ-field generated by $\{\varepsilon_{t-s}; s \geqslant i\}$ where (ε_t) is a nonanticipative solution of model* (1). *Then*

$$\begin{aligned}\pi(k)E[X_{t-i}|\Delta_t = k] &= \sum_{j=1}^{d} E[X_{t-i}|\Delta_{t-1} = j]p(j,k)\pi(j)\\ &= \sum_{j=1}^{d} E[X_{t-i}|\Delta_{t-i} = j]p^{(i)}(j,k)\pi(j).\end{aligned}$$

Proof. First, note that for any measurable partition (Ω_j) of Ω, the expectation of any integrable variable can be decomposed as $E(X) = \sum_j E(X|\Omega_j)P(\Omega_j)$ (see e.g. [4, 3.4.1 and 3.4.2]). Hence we deduce that

$$E[X_{t-i}|\Delta_t = k] = \sum_{j=1}^{d} E[X_{t-i}|\Delta_t = k, \Delta_{t-1} = j]P[\Delta_{t-1} = j|\Delta_t = k].$$

Now observe that for all j's such that $P[\Delta_t = k, \Delta_{t-1} = j] > 0$, and for all x

$$\begin{aligned}&P[X_{t-i} < x|\Delta_t = k, \Delta_{t-1} = j]\\ &\quad = \frac{P[\Delta_t = k|\Delta_{t-1} = j, X_{t-i} < x]P[X_{t-i} < x, \Delta_{t-1} = j]}{P[\Delta_t = k, \Delta_{t-1} = j]}\\ &\quad = \frac{P[\Delta_t = k|\Delta_{t-1} = j]P[X_{t-i} < x|\Delta_{t-1} = j]P[\Delta_{t-1} = j]}{P[\Delta_t = k, \Delta_{t-1} = j]}\\ &\quad = P[X_{t-i} < x|\Delta_{t-1} = j].\end{aligned}$$

Hence $E[X_{t-i}|\Delta_t = k, \Delta_{t-1} = j] = E[X_{t-i}|\Delta_{t-1} = j]$. In view of $P[\Delta_{t-1} = j|\Delta_t = k]\pi(k) = P[\Delta_t = k|\Delta_{t-1} = j]\pi(j)$, the first equality follows. The second equality can be shown similarly. □

Proof (*Proof of Theorem 2*). The proof is split into three steps:

(i) We proceed by evaluating the expectation of $\varepsilon_t^2\varepsilon_{t-\ell}^2$ conditional to the regime of Δ_t. For $\ell > 0$ and $k \in \mathscr{E}$ we have

$$\begin{aligned}E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t = k] &= E[h_t\varepsilon_{t-\ell}^2|\Delta_t = k]\\ &= \omega(k)E[\varepsilon_{t-\ell}^2|\Delta_t = k] + \sum_{i=1}^{q} \alpha_i(k)E[\varepsilon_{t-i}^2\varepsilon_{t-\ell}^2|\Delta_t = k]\\ &\quad + \sum_{j=1}^{p} \beta_j(k)E[h_{t-j}\varepsilon_{t-\ell}^2|\Delta_t = k].\end{aligned}$$

Moreover, for $\ell > j$ we have $E[\varepsilon_{t-j}^2\varepsilon_{t-\ell}^2|\Delta_t = k] = E[h_{t-j}\varepsilon_{t-\ell}^2|\Delta_t = k]$, so that

$$E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t = k] = \omega(k)E[\varepsilon_{t-\ell}^2|\Delta_t = k] + \sum_{i=1}^{r} \xi_i(k)E[\varepsilon_{t-i}^2\varepsilon_{t-\ell}^2|\Delta_t = k], \quad \ell > p.$$

In view of Lemma 3, we have, for $\ell > p$ and $\ell \geqslant q$

$$\pi(k)E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t = k] = \pi(k)\omega(k)E[\varepsilon_{t-\ell}^2|\Delta_t = k] + \sum_{i=1}^{r}\sum_{j=1}^{d} E[\varepsilon_{t-i}^2\varepsilon_{t-\ell}^2|\Delta_{t-i} = j]\xi_i(k)p^{(i)}(j,k)\pi(j)$$

which, by the notations in (2), can be written in vector form as

$$\Pi_{E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t=\cdot]} = \Pi_{\omega(\cdot)E[\varepsilon_{t-\ell}^2|\Delta_t=\cdot]} + \sum_{i=1}^{r} \mathbb{P}_{\xi_i}^{(i)}\Pi_{E[\varepsilon_t^2\varepsilon_{t-\ell+i}^2|\Delta_t=\cdot]}, \quad \ell \geqslant \max\{p+1, q\}. \tag{21}$$

Further we have, using Lemma 3

$$\pi(k)E[\varepsilon_{t-\ell}^2|\Delta_t = k] = \sum_{j=1}^{d} E[\varepsilon_{t-\ell}^2|\Delta_{t-1} = j]p(j,k)\pi(j), \quad \ell > 0$$

or equivalently

$$\Pi_{E[\varepsilon_{t-\ell}^2|\Delta_t=\cdot]} = \mathbb{P}\Pi_{E[\varepsilon_{t-\ell+1}^2|\Delta_t=\cdot]}, \quad \ell > 0. \tag{22}$$

Similarly

$$\Pi_{\omega(\cdot)E[\varepsilon_{t-\ell}^2|\Delta_t=\cdot]} = \mathbb{P}_\omega\Pi_{E[\varepsilon_{t-\ell+1}^2|\Delta_t=\cdot]}, \quad \ell > 0. \tag{23}$$

This leads us to introduce the $d(r+1)\times 1$ vector

$$W(\ell) := \begin{pmatrix} \Pi_{E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t=\cdot]} \\ \Pi_{E[\varepsilon_t^2\varepsilon_{t-\ell+1}^2|\Delta_t=\cdot]} \\ \vdots \\ \Pi_{E[\varepsilon_t^2\varepsilon_{t-\ell+r-1}^2|\Delta_t=\cdot]} \\ \Pi_{E[\varepsilon_{t-\ell}^2|\Delta_t=\cdot]} \end{pmatrix}, \quad \ell = 0, \pm 1, \ldots .$$

From (21)–(23) we have

$$W(\ell) = AW(\ell-1), \quad \forall \ell > \max\{p+1, q\} - 1. \tag{24}$$

It should be noted that, because the components of $W(\ell)$ are positive, and bounded when ℓ increases by stationarity of (ε_t^2), (24) implies that $\rho(A) \leqslant 1$. The autocovariance function of (ε_t^2) is given by

$$\gamma_{\varepsilon^2}(\ell) = \sum_{k=1}^{d} E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t = k]\pi(k) - \gamma_\varepsilon^2(0),$$

where $\gamma_\varepsilon(0) = E(\varepsilon_t^2)$. In this equality, the right-hand terms can be expressed as

$$\gamma_\varepsilon(0) = \underline{1}_d'\Pi_{E[\varepsilon_{t-\ell-r+1}^2|\Delta_t=\cdot]} \quad \text{and} \quad \sum_{k=1}^{d} E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t = k]\pi(k) = \underline{1}_d'\Pi_{E[\varepsilon_t^2\varepsilon_{t-\ell}^2|\Delta_t=\cdot]},$$

where $\underline{1}_d = (1,\dots,1)'$ denotes the d element sum vector. We therefore find that, in view of (24)

$$\gamma_{\varepsilon^2}(\ell) = \delta' W(\ell + r - 1), \quad \forall \ell,$$
$$W(\ell + r - 1) = A W(\ell + r - 2), \quad \forall \ell > \max\{p+1, q\} - r \tag{25}$$

with $\delta' = (0,\dots,0,\underline{1}'_d, -\underline{1}'_d\gamma_\varepsilon(0))$. By the first part of Lemma 2 with $n_0 = n_0(A)$, $n = d(r+1)$ and $\mathscr{P} = \mathscr{P}_A$, we find that (ε_t^2) admits an ARMA representation, with orders $(d(r+1) - n_0(A), d(r+1))$ when $p \geqslant q$, and $(d(r+1) - n_0(A), d(r+1) - 1)$ when $p < q$.

(ii) To reduce the ARMA orders, in view of the second part of Lemma 2, we now show that $X_t := \varepsilon_t^2 - E(\varepsilon_t^2)$ is purely nondeterministic. It is sufficient to show that

$$E|X_t - P_{\mathscr{H}_{t-h}(X)} X_t|^2 \to EX_t^2 \quad \text{when } h \to \infty,$$

where $P_{\mathscr{H}_n(X)}$ denotes the projection mapping onto $\mathscr{H}_n(X)$. This amounts to showing that $\mathrm{Var}(P_{\mathscr{H}_{t-h}(X)} X_t)$ tends to zero as $h \to \infty$. From

$$\begin{aligned}\mathrm{Var}(X_t) &= \mathrm{Var}(P_{\mathscr{H}_{t-h}(X)} X_t) + \mathrm{Var}(X_t - P_{\mathscr{H}_{t-h}(X)} X_t) \\ &= \mathrm{Var}\{E(X_t | X_{t-s}, s \geqslant h)\} + \mathrm{Var}\{X_t - E(X_t | X_{t-s}, s \geqslant h)\}\end{aligned}$$

we deduce that $\mathrm{Var}\{E(X_t|X_{t-s}, s\geqslant h)\} \geqslant \mathrm{Var}(P_{\mathscr{H}_{t-h}(X)} X_t)$. By the same argument we have $\mathrm{Var}\{E(X_t|\Delta_{t-s}, \eta_{t-s}, s\geqslant h)\} \geqslant \mathrm{Var}\{E(X_t|X_{t-s}, s\geqslant h)\}$. It is therefore sufficient to show that $\|\mathrm{Var}\{E(\underline{z}_t^*|\Delta_{t-s}, \eta_{t-s}, s\geqslant h)\}\|$ tends to zero as $h \to \infty$, where $\underline{z}_t^*$ is defined in (7). Using Lemma 1, we obtain

$$\begin{aligned}E(\underline{z}_t^*|\Delta_{t-s}, \eta_{t-s}, s\geqslant h) = {} & \mathbb{I}\mathbb{P}^{(h)}_{\underline{b}^{(1)}} e_{\Delta_{t-h}} + \sum_{k=1}^{h-1} \mathbb{I}\mathbb{P}^{k}_{C^{(1)}} \mathbb{P}^{(h-k)}_{\underline{b}^{(1)}} e_{\Delta_{t-h}} \\ & + \sum_{k=h}^{\infty} \mathbb{I}\mathbb{P}^{h}_{C^{(1)}} \{e_{\Delta_{t-h}} \otimes I_{p+q}\} C_{t-h} \cdots C_{t-k+1} \underline{b}_{t-k},\end{aligned}$$

where $\mathbb{I} = \underline{1}_d \otimes I_{p+q}$. Using the Minkovski inequality, $\|\mathbb{P}^{h}_{C^{(1)}}\| \to 0$ and

$$\mathbb{P}^{(h)}_{\underline{b}^{(1)}} e_i \to \Pi_{\underline{b}^{(1)}} \quad \text{as } h \to \infty, \text{ for all } i = 1,\dots,d$$

it is easy to show that $\|E(\underline{z}_t^*|\Delta_{t-s}, \eta_{t-s}, s\geqslant h) - E(\underline{z}_t^*)\|_2 \to 0$, as $h \to \infty$. We conclude that (X_t) is purely nondeterministic. Applying the second part of Lemma 2, we find that ε_t^2 has an ARMA representation, with orders $(N, d(r+1) - n_1(A))$ when $p \geqslant q$, and $(N, d(r+1) - n_1(A) - 1)$ when $p < q$. The AR order is the announced result. When $n_0(A) = 0$ we get the desired MA order.

(iii) The MA order can be further reduced when $n_0(A) \neq 0$. We will show that (24) and (25) hold, where $W(\cdot)$, δ and A are replaced by matrices of smaller sizes, $\tilde{W}(\cdot)$, $\tilde{\delta}$ and $\tilde{A}$, with $\tilde{A}$ non-singular. There exist a full rank matrix $\mathbb{P}^*$, obtained by cancelling $n_0(\mathbb{P})$ redundant columns of $\mathbb{P}$, and a full rank $\{d - n_0(\mathbb{P})\} \times d$ matrix $\Lambda_{\mathbb{P}}$ such that $\mathbb{P} = \mathbb{P}^* \Lambda_{\mathbb{P}}$. Because $\omega(\cdot) > 0$ we also have $\mathbb{P}_\omega = \mathbb{P}^*_\omega \Lambda_{\mathbb{P}}$ for some full rank matrix $\mathbb{P}^*_\omega$. We similarly introduce a full rank matrix $\mathbb{P}^{*(r)}$ and a full rank $\{d - n_0(\mathbb{P}^{(r)}_{\xi_r})\} \times d$ matrix $\Lambda_{\mathbb{P}^{(r)}}$ such that $\mathbb{P}^{(r)}_{\xi_r} = \mathbb{P}^{*(r)} \Lambda_{\mathbb{P}^{(r)}}$. Now define the block-diagonal $\{d(r+1) - n_0(A)\} \times d(r+1)$ matrix $\Lambda_A = \mathrm{diag}(I_{d(r-1)}, \Lambda_{\mathbb{P}^{(r)}}, \Lambda_{\mathbb{P}})$, and let

the $d(r+1)\times\{d(r+1)-n_0(A)\}$ matrix A^* such that $A=A^*\Lambda_A$. Let $\tilde{A}=\Lambda_A A^*$. Remark that A^*, Λ_A and $\tilde{A}$ are full-rank matrices, and that the square matrices A and $\tilde{A}$ have the same rank (when $n_0(A)>0$, A is not full-rank of course). We have by (25)

$$\tilde{W}(\ell+r-1):=\Lambda_A W(\ell+r-1)=\tilde{A}\tilde{W}(\ell+r-2),\quad \forall \ell>\max\{p+1,q\}-r.$$

Note that $\underline{1}'_d=\underline{1}'_d\mathbb{P}=\underline{1}'_d\mathbb{P}^*\Lambda_{\mathbb{P}}=\underline{1}'_{d-n_0(\mathbb{P})}\Lambda_{\mathbb{P}}=\underline{1}'_{d-n_0(\mathbb{P}^{(r)}_{\xi_r})}\Lambda_{\mathbb{P}^{(r)}}$. Thus $\delta'=\tilde{\delta}'\Lambda_A$ with $\tilde{\delta}'=(0,\ldots,0,\underline{1}'_{d-n_0(\mathbb{P}^{(r)}_{\xi_r})},-\underline{1}'_{d-n_0(\mathbb{P})}\gamma_\varepsilon(0))$. Hence $\gamma_{\varepsilon^2}(\ell)=\tilde{\delta}'\tilde{W}(\ell+r-1)$. Applying Lemma 2, with $n_0=0$ (because $\tilde{A}$ has full rank), $n=d(r+1)-n_0(A)$ and $\mathscr{P}=\mathscr{P}_{\tilde{A}}$ (the characteristic polynomial of $\tilde{A}$), we find that (ε_t^2) admits an ARMA representation, with orders $(d(r+1)-n_0(A),d(r+1)-n_0(A))$ when $p\geqslant q$, and $(d(r+1)-n_0(A),d(r+1)-n_0(A)-1)$ when $p<q$. It is easy to show that A and $\tilde{A}$ have the same non-zero eigenvalues. By cancelling the roots of modulus 1, as in step (ii), we obtain the result of the theorem and the conclusion of Remark 7. □

Illustrations of this theorem, as well as of Theorem 3 below, are provided in an extended version of this paper [15].

4. Autocovariance structure of (ε_t^{2m})

In this section we extend the results of the previous section to powers of (ε_t^2). For statistical purposes, such as identification, considering only the ARMA representation for ε_t^2 will not suffice. Indeed, two different models of the form (1) may lead to the same ARMA representation for the squares, but, in general, higher powers will have different representations.

We will establish that for any positive integer m, the process (ε_t^{2m}) is, under mild assumptions, an ARMA. This result is much harder to prove than for (ε_t^2). Before dealing with the general case, we proceed with the case $p=q=1$, which can be handled in a simpler way and provides sharper results as far as the ARMA orders are concerned. The $2m$th order moment of the iid process is denoted by $\mu_{2m}=E(\eta_t^{2m})$.

4.1. The case $p=q=1$

Let

$$\xi_{m_1,m_2}(\cdot)=\binom{m_1}{m_2}\{\omega(\cdot)\}^{m_1-m_2}E_\eta[\alpha(\cdot)\eta_{t-1}^2+\beta(\cdot)]^{m_2}\frac{\mu_{2m_1}}{\mu_{2m_2}}. \tag{26}$$

Note that $\xi_{1,1}(\cdot)=\xi(\cdot)$, as defined in (17). This notation is introduced to establish formulas of the form $E[\varepsilon_t^{2m_1}|\Delta_t]=\sum_{m_2=0}^{m_1}\xi_{m_1,m_2}(\Delta_t)E[\varepsilon_{t-1}^{2m_2}|\Delta_t]$, for $m_1\leqslant 2m$. Let

the $d(m+1)\times d(m+1)$ matrix

$$A_m=\begin{pmatrix} \mathbb{P}_{\xi_{m,m}} & \mathbb{P}_{\xi_{m,m-1}} & \cdots & \cdots & \mathbb{P}_{\xi_{m,0}} \\ 0 & \mathbb{P}_{\xi_{m-1,m-1}} & \cdots & \cdots & \mathbb{P}_{\xi_{m-1,0}} \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \cdots & 0 & \mathbb{P}_{\xi_{1,1}} & \mathbb{P}_{\xi_{1,0}} \\ 0 & \cdots & 0 & 0 & \mathbb{P} \end{pmatrix}.$$

Theorem 3. *Let $p=q=1$ and suppose $\alpha(\cdot)+\beta(\cdot)>0$. Suppose that $\rho[\mathbb{P}_{\xi_{2m,2m}}]<1$ and let (ε_t) be the nonanticipative 4mth-order stationary solution of model* (1). *Then (ε_t^{2m}) is an* ARMA(N_1,N_1) *process, with $N_1:=(m+1)\{d-n_0(\mathbb{P})\}-n_1(A_m)$.*

Remark 9. Recall that by Corollary 2, $\rho[\mathbb{P}_{\xi_{2m,2m}}]<1$ is a necessary and sufficient condition for the existence of $E\varepsilon_t^{4m}$. In view of (3) we have

$$\det(A_m)=\{\det(\mathbb{P})\}^{m+1}\prod_{m_1=1}^{m}\prod_{k=1}^{d}\xi_{m_1,m_1}(k),\quad \text{with } \xi_{m_1,m_1}(\cdot)=E_\eta\{\alpha(\cdot)\eta_{t-1}^2+\beta(\cdot)\}^{m_1}. \tag{27}$$

Notice that, since $\xi_{m_1,m_1}(\cdot)>0$, we have $n_0(A_m)=(m+1)n_0(\mathbb{P})$. The characteristic polynomial of A_m is therefore of the form

$$\mathscr{P}_{A_m}(x)=x^{(m+1)n_0(\mathbb{P})}\prod_{i=1}^{n_1(A_m)}(x-\lambda_i)\mathscr{P}_2(x),\quad \mathscr{P}_2(0)\neq 0, \tag{28}$$

where the λ_i are the eigenvalues of modulus 1 of A_m, and $\mathscr{P}_2$ is a polynomial whose roots are inside the unit circle. It will be shown below that the AR polynomial of the ARMA model is $L^{(m+1)\{d-n_0(\mathbb{P})\}-n_1(A_m)}\mathscr{P}_2(L^{-1})$.

Remark 10. In the special case when $d=1$ we have $\alpha(\cdot)=\alpha$, $\beta(\cdot)=\beta$, $\mathbb{P}=1$ and $n_1(A_m)=1$. Therefore we find that for the standard GARCH$(1,1)$ model, the process (ε_t^{2m}) is an ARMA(m,m). This result is well-known when $m=1$ and, to our knowledge, had not been proved for higher exponents m. There is an important difference, however, between the case $m=1$ and the general case. When $m=1$, the ARMA representation is a strong one in the sense that the noise is a martingale difference. This will generally not be the case for higher exponents m.

Remark 11. The AR polynomial of the ARMA representation for (ε_t^{2m}) is the product of the AR polynomial of the ARMA representation for $(\varepsilon_t^{2(m-1)})$, and a polynomial obtained from the matrix $\mathbb{P}_{\xi_{m,m}}$. In particular, the AR part of the representation does not depend on $\omega(\cdot)$, because $\xi_{m_1,m_1}(k)=E\{\alpha(k)\eta_{t-1}^2+\beta(k)\}^{m_1}$, and hence the block-diagonal terms of A_m, are independent of $\omega(\cdot)$. Note also that Remark 8 remains valid.

Before proving the theorem, we will establish a basic result we will use to investigate the autocovariance structure of (ε_t^{2m}).

Lemma 4. *Under the assumptions of Theorem* 3 *and Lemma* 3, *if* $j<i$, *we have for any* $m_1 \in \{0,1,\ldots,m\}$

$$E[\varepsilon_{t-j}^{2m_1} X_{t-i}|\Delta_t = k] = \mu_{2m_1} E[h_{t-j}^{m_1} X_{t-i}|\Delta_t = k]$$

and, for nonnegative integers m_1 *and* m_2 *such that* $m_1 + m_2 \leqslant m$

$$E[\varepsilon_{t-j}^{2m_1} h_{t-j}^{m_2} X_{t-i}|\Delta_t = k] = \frac{\mu_{2m_1}}{\mu_{2(m_1+m_2)}} E[\varepsilon_{t-j}^{2(m_1+m_2)} X_{t-i}|\Delta_t = k].$$

Proof. The first statement is a trivial consequence of the equality $\varepsilon_{t-j} = \sqrt{h_{t-j}}\eta_{t-j}$ and the independence between η_{t-j} and all other variables involved in the conditional expectation. To prove the second statement, write $E[\varepsilon_{t-j}^{2m_1} h_{t-j}^{m_2} X_{t-i}|\Delta_t = k] = E[\eta_{t-j}^{2m_1} h_{t-j}^{m_1+m_2} X_{t-i}|\Delta_t = k] = \mu_{2m_1} E[h_{t-j}^{m_1+m_2} X_{t-i}|\Delta_t = k]$ and use the first part of the lemma. □

Proof (*Proof of Theorem 3*). Observe that for $\ell>0$, and for $m_1 \in \{0,1,\ldots,m\}$, $k \in \mathscr{E}$

$$\begin{aligned} &E[\varepsilon_t^{2m_1} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \\ &\quad = \mu_{2m_1} \sum_{m_2=0}^{m_1} \binom{m_1}{m_2} E[\{\omega(\Delta_t)\}^{m_1-m_2} \{\alpha(\Delta_t)\varepsilon_{t-1}^2 + \beta(\Delta_t)h_{t-1}\}^{m_2} \varepsilon_{t-\ell}^{2m}|\Delta_t = k]. \end{aligned} \quad (29)$$

Moreover, for $\ell>1$

$$\begin{aligned} &E[\{\alpha(\Delta_t)\varepsilon_{t-1}^2 + \beta(\Delta_t)h_{t-1}\}^{m_2} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \\ &\quad = E[\{\alpha(\Delta_t)\eta_{t-1}^2 + \beta(\Delta_t)\}^{m_2} h_{t-1}^{m_2} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \\ &\quad = E\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m_2} E[h_{t-1}^{m_2} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \\ &\quad = E\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m_2} \frac{1}{\mu_{2m_2}} E[\varepsilon_{t-1}^{2m_2} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \end{aligned}$$

by Lemma 4. Hence, in view of (29) and Lemma 3, we have for $\ell>1$

$$\begin{aligned} \pi(k)E[\varepsilon_t^{2m_1} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] &= \pi(k) \sum_{m_2=0}^{m_1} \xi_{m_1,m_2}(k) E[\varepsilon_{t-1}^{2m_2} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \\ &= \sum_{m_2=0}^{m_1} \sum_{j=1}^{d} \xi_{m_1,m_2}(k) E[\varepsilon_{t-1}^{2m_2} \varepsilon_{t-\ell}^{2m}|\Delta_{t-1} = j] p(j,k)\pi(j) \end{aligned} \quad (30)$$

which can be written in vector form as

$$\Pi_{E[\varepsilon_t^{2m_1} \varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} = \sum_{m_2=0}^{m_1} \mathbb{P}_{\xi_{m_1,m_2}} \Pi_{E[\varepsilon_t^{2m_2} \varepsilon_{t-\ell+1}^{2m}|\Delta_t=\cdot]} \quad \text{for all } \ell>1. \quad (31)$$

Because (31) holds for all $m_1 \in \{0, 1, \ldots, m\}$, we have for $\ell > 1$

$$W_m(\ell) = A_m W_m(\ell - 1) \quad \text{where } W_m(\ell) = \begin{pmatrix} \Pi_{E[\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m}|\Lambda_t=\cdot]} \\ \Pi_{E[\varepsilon_t^{2(m-1)}\varepsilon_{t-\ell}^{2m}|\Lambda_t=\cdot]} \\ \vdots \\ \vdots \\ \Pi_{E[\varepsilon_{t-\ell}^{2m}|\Lambda_t=\cdot]} \end{pmatrix}.$$

Moreover,

$$\gamma_{\varepsilon^{2m}}(\ell) = \underline{1}_d' \Pi_{E[\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m}|\Lambda_t=\cdot]} - E(\varepsilon_t^{2m})\underline{1}_d' \Pi_{E[\varepsilon_{t-\ell}^{2m}|\Lambda_t=\cdot]} = v' W_m(\ell) \tag{32}$$

with $v' = (\underline{1}_d', 0, \ldots, 0, -E(\varepsilon_t^{2m})\underline{1}_d')$.

Hence, by Lemma 2, with $n_0 = n_0(A_m)$, $\ell_0 = 1$, $n = d(m+1)$ and $\mathscr{P} = \mathscr{P}_{A_m}$, we find that (ε_t^{2m}) admits an ARMA$(d(m+1) - n_0(A_m), d(m+1))$ representation.

With the notations of the proof of Theorem 2, we introduce the full-rank block-diagonal $(m+1)\{d - n_0(\mathbb{P})\} \times d(m+1)$ matrix $\Lambda_{A_m} = \operatorname{diag}(\Lambda_{\mathbb{P}}, \ldots, \Lambda_{\mathbb{P}})$, and the matrices A_m^* and $\tilde{A}_m$ such that $A_m = A_m^* \Lambda_{A_m}$, and $\tilde{A}_m = \Lambda_{A_m} A_m^*$ is a non singular $(m+1)\{d - n_0(\mathbb{P})\} \times (m+1)\{d - n_0(\mathbb{P})\}$ matrix. We have

$$\tilde{W}_m(\ell) := \Lambda_{A_m} W_m(\ell) = \tilde{A}_m \tilde{W}_m(\ell - 1), \quad \forall \ell > 1.$$

Thus $v' = \tilde{v}' \Lambda_{A_m}$ with $\tilde{v}' = (\underline{1}_{d-n_0(\mathbb{P})}', 0, \ldots, 0, -E(\varepsilon_t^{2m})\underline{1}_{d-n_0(\mathbb{P})}')$. Hence $\gamma_{\varepsilon^{2m}}(\ell) = \tilde{v}' \tilde{W}_m(\ell)$. Applying the first part of Lemma 2, with $n = (m+1)\{d - n_0(\mathbb{P})\}$, $n_0(\tilde{A}_m) = 0$, $\ell_0 = 1$, we may conclude that ε_t^{2m} is an ARMA$((m+1)\{d - n_0(\mathbb{P})\}, (m+1)\{d - n_0(\mathbb{P})\})$ process.

An argument similar to that used in Step (ii) of the proof of Theorem 2 shows that $\varepsilon_t^{2m} - E(\varepsilon_t^{2m})$ is purely nondeterministic. The proof is completed by applying the second part of Lemma 2. □

4.2. The general case

We start this section by briefly describing the scheme of proof and by introducing some new notation. For ease of presentation we first assume that $p > 1$ and $q > 1$ but the theorem of this section will be stated for $p \geqslant 1$, $q \geqslant 1$. Appropriate extensions of the notations will be given in Remark 14 below.

Proceeding as in Section 4.1, we will compute moments of the process (ε_t) conditional on the Markov chain. However, instead of moments of variables of the form $\varepsilon_t^{2m_1}\varepsilon_{t-\ell}^{2m}$ as in the case $p = q = 1$, it will be necessary to consider variables of the form $\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}$ where $\Upsilon_t(\underline{m}')$ is defined by

$$\Upsilon_t(\underline{m}') = \varepsilon_t^{2m_0} \ldots \varepsilon_{t-q+1}^{2m_{q-1}} h_{t-1}^{n_1} \ldots h_{t-p+1}^{n_{p-1}},$$
$$\text{for } \underline{m}' = (m_0, \ldots, m_{q-1}, n_1, \ldots, n_{p-1}) \in \mathbb{N}^{p+q-1}. \tag{33}$$

$$\mathcal{R}_{m'} = \{\underline{m}' \in \mathbb{N}^{p+q-1} \mid m_0 + \cdots + m_{q-1} + n_1 + \cdots + n_{p-1} = m'\},$$

$$\underline{m}' \to (m_0,\ m_1,\ \ldots,\ m_{q-1},\ 0,\ n_1,\ \ldots,\ n_{p-1},\ 0)$$

$$\underline{l} = (l_0,\ l_1,\ \ldots,\ l_{q-1},\ l_q,\ l_{q+1},\ \ldots,\ l_{p+q-1},\ l_{p+q})$$

$$\downarrow \text{ sum}$$

$$(l_0 + m_0, \boxed{l_1 + m_1}, \ldots, l_{q-1} + m_{q-1}, l_q, \boxed{l_{q+1} + n_1}, \ldots, l_{p+q-1} + n_{p-1}, l_{p+q})$$

$$\downarrow \text{ cancellation of the 1-st element, summation of the boxed terms}$$

$$(\boxed{l_1 + m_1 + l_{q+1} + n_1}, l_2 + m_2, \ldots, l_{q-1} + m_{q-1}, l_q, l_{q+2} + n_2, \ldots, l_{p+q-1} + n_{p-1}, l_{p+q})$$

$$= \underline{m}' \star \underline{l}$$

Fig. 1. Operator $\star$.

To compute the expectation of $\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}$ conditional on $\Delta_t = k$, we will replace ε_t^2 by $h_t\eta_t^2$ and apply the multinomial formula to $h_t^{m_0}$, with h_t expressed in function of the ε_{t-i}^2 and h_{t-j}. It will therefore be convenient to introduce the real numbers

$$\xi_{\underline{m}',\underline{l}}(k) = \frac{m_0!}{\Pi_{j=0}^{p+q} l_j!}\omega(k)^{l_0}\alpha_1(k)^{l_1}\ldots\alpha_q(k)^{l_q}\beta_1(k)^{l_{q+1}}\ldots\beta_p(k)^{l_{p+q}}\frac{\mu_{2m_0}\mu_{2(l_1+m_1)}}{\mu_{2(l_1+m_1+l_{q+1}+n_1)}}$$

defined for any $k \in \mathscr{E}$, $\underline{l} = (l_0, \ldots, l_{p+q}) \in \mathbb{N}^{p+q+1}$ and $m_0, m_1 \in \mathbb{N}$. Thus we will obtain a formula relying on the expectation of $\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}$ conditional on $\Delta_t = k$, and the expectations, conditional on $\Delta_{t-1} = j$, of terms of the form

$$\varepsilon_{t-1}^{2(l_1+m_1+l_{q+1}+n_1)}\ldots\varepsilon_{t-q}^{2(l_q+m_q)}h_{t-2}^{l_{q+2}+n_2}\ldots h_{t-p}^{l_{p+q}+n_p}\varepsilon_{t-\ell}^{2m} := \Upsilon_{t-1}(\underline{m}' \star \underline{l})\varepsilon_{t-\ell}^{2m} \tag{34}$$

with for notational convenience $m_q = n_p = 0$. This formula involves an operator, $\star$, from $\mathbb{N}^{p+q-1} \times \mathbb{N}^{p+q+1}$ to $\mathbb{N}^{p+q-1}$ defined, for $p \geqslant 2$ and $q \geqslant 2$, by

$$\underline{m}' \star \underline{l} = (l_1 + m_1 + l_{q+1} + n_1, l_2 + m_2, \ldots, l_q + m_q, l_{q+2} + n_2, \ldots, l_{p+q} + n_p),$$

where $\underline{m}'$ is defined as above and $\underline{l} = (l_0, l_1, \ldots, l_{p+q})$. For convenience, the operation carried out by $\star$ is displayed in Fig. 1.

Because we are interested in the autocovariance structure of ε_t^{2m}, we will focus on variables $\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}$ such that the sum of the exponents in the right-hand side of (33) is less than or equal to $2m$. Similarly, having applied the multinomial formula to $h_t^{m_0}$, the sum of the elements of $\underline{l}$ in (34) is equal to m_0. We will therefore introduce simplexes on $\mathbb{N}^{p+q-1}$ and $\mathbb{N}^{p+q+1}$ defined, for $m' \leqslant m$ by

$$\mathscr{R}_{m'} = \{\underline{m}' \in \mathbb{N}^{p+q-1} \mid m_0 + \cdots + m_{q-1} + n_1 + \cdots + n_{p-1} = m'\}$$

with by convention $n_1 + \cdots + n_{p-1} = 0$ if $p = 1$, and for any positive integer $m_0 \leqslant m$, by

$$\mathscr{S}_{m_0} = \{\underline{l} = (l_0, \ldots, l_{p+q}) \in \mathbb{N}^{p+q+1} \mid l_0 + \cdots + l_{p+q} = m_0\}.$$

Denote the first component of $\underline{m}'$ by $m_0(\underline{m}')$. We now order the elements of $\widetilde{\mathscr{R}}_m := \bigcup_{m'=0}^{m} \mathscr{R}_{m'}$ in the following way: if $\underline{m}' \in \mathscr{R}_{m'}$ and $\underline{m}'' \in \mathscr{R}_{m''}$ with $m' > m''$ then we say that $\underline{m}' > \underline{m}''$; for the elements of $\mathscr{R}_{m'}$ we take the so-called lexicographical order. Let $\tau(\underline{m}')$ denote the rank of any element $\underline{m}'$ in $\widetilde{\mathscr{R}}_m$.For instance $\tau(m,0,\ldots,0) = |\widetilde{\mathscr{R}}_m|$, where $|A|$ denotes the cardinal of any finite set A, and $\tau(0,\ldots,0) = 1$. Let

$$\mathscr{A}_m = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1\widetilde{|\mathscr{R}_m|}} \\ A_{21} & A_{22} & \cdots & A_{2\widetilde{|\mathscr{R}_m|}} \\ \vdots & \ddots & & \vdots \\ A_{\widetilde{|\mathscr{R}_m|}-1,1} & \cdots & & A_{\widetilde{|\mathscr{R}_m|}-1,\widetilde{|\mathscr{R}_m|}} \\ 0 & \cdots & 0 & \mathbb{P} \end{pmatrix}, \tag{35}$$

where

$$A_{ij} = \sum_{\underline{l} \in \mathscr{L}_{i,j}} \mathbb{P}_{\xi_{\widetilde{\tau^{-1}\{|\mathscr{R}_m|-i+1\},\underline{l}}}} = \mathbb{P}_{\sum_{\underline{l} \in \mathscr{L}_{i,j}} \xi_{\widetilde{\tau^{-1}\{|\mathscr{R}_m|-i+1\},\underline{l}}}}$$

and $\mathscr{L}_{i,j} = \{\underline{l} \in \mathscr{S}_{m_0\{\widetilde{\tau^{-1}(|\mathscr{R}_m|-i+1)}\}} \mid \tau\{\tau^{-1}(|\widetilde{\mathscr{R}}_m| - i + 1) \star \underline{l}\} = |\widetilde{\mathscr{R}}_m| - j + 1\}$.

Remark 12. It can be seen that the sum of the elements of $\underline{m}' \star \underline{l}$ is always less than that of the elements of $\underline{m}'$ when $\underline{l} \in \mathscr{S}_{m_0(\underline{m}')}$. So $A_{ij} = 0$ if $i > |\mathscr{R}_m| + \cdots + |\mathscr{R}_{m-k}|$ and $j \leqslant |\mathscr{R}_m| + \cdots + |\mathscr{R}_{m-k}|$, for any $k \geqslant 0$. The matrix $\mathscr{A}_m$ is therefore upper block-triangular, with blocks of size $d|\mathscr{R}_m|, \ldots, d|\mathscr{R}_1|$ and $d|\mathscr{R}_0| = d$ on the diagonal.

Remark 13. From $n^{(k)} = n(n-1)\ldots(n-k+1)$ and the formula $\sum_{n=k}^{N} n^{(k)} = \frac{(N+1)^{(k+1)}}{k+1}$ we have

$$\begin{aligned} |\widetilde{\mathscr{R}}_m| &= \sum_{m'=0}^{m} |\mathscr{R}_{m'}| = \sum_{m'=0}^{m} \frac{(m'+p+q-2)^{(p+q-2)}}{(p+q-2)!} \\ &= \frac{(m+p+q-1)^{(p+q-1)}}{(p+q-1)!} = \begin{pmatrix} m+p+q-1 \\ p+q-1 \end{pmatrix}. \end{aligned}$$

Remark 14. The previous notations can be extended as follows to the case $p = 1$ or $q = 1$. The definition of $\Upsilon_t(\underline{m}')$ in (33) remains valid for $q = 1$, and it can be extended to the case $p = 1$ with the following convention, which we will use throughout: $X_1^{a_1} \ldots X_n^{a_n} = 1$ if $n < 1$. For instance, if $p = q = 1$ we have $\mathscr{R}_{m'} = \{m'\}$ and $\Upsilon_t(\underline{m}') = \Upsilon_t(m') = \varepsilon_t^{2m'}$. The definition of the operator $\star$ is extended as follows, for $p = 1$ and $q \geqslant 2$, $\underline{m}' \star \underline{l} := (l_1 + m_1 + l_{q+1}, l_2 + m_2, \ldots, l_q + m_q) \in \mathbb{N}^q$; for $p \geqslant 2$ and $q = 1$, $\underline{m}' \star \underline{l} := (l_1 + l_2 + n_1, l_3 + n_2, \ldots, l_{p+1} + n_p) \in \mathbb{N}^p$; for $p = 1$ and $q = 1$, $\underline{m}' \star \underline{l} = l_1 + l_2$.

To illustrate these notations, consider the following example.

Example 4.1. Let $p=1$, $q=2$ and $m=2$. Then $\widetilde{\mathscr{R}}_2$ has 6 elements: $(2,0)>(1,1)>(0,2)>(1,0)>(0,1)>(0,0)$. Hence $\tau(0,0)=1, \tau(0,1)=2, \tau(1,0)=3, \tau(0,2)=4, \tau(1,1)=5, \tau(2,0)=6$. For instance, let us compute A_{11}. We have

$$m_0\{\tau^{-1}(6)\}=m_0\{(2,0)\}=2, \quad \mathscr{S}_2=\{(2,0,0,0),(0,2,0,0),\ldots,(0,0,1,1)\}.$$

Moreover, $(2,0)\star(l_0,l_1,l_2,l_3)=(l_1+l_3,l_2)$. Hence

$$\mathscr{L}_{1,1}=\{\underline{l}\in\mathscr{S}_2|\tau\{(l_1+l_3,l_2)\}=6\}=\{(0,2,0,0),(0,0,0,2),(0,1,0,1)\}.$$

Therefore, we compute

$$\xi_{(2,0),(0,2,0,0)}(\cdot)=\mu_4\alpha_1^2(\cdot), \quad \xi_{(2,0),(0,0,0,2)}(\cdot)=\beta_1^2(\cdot), \quad \xi_{(2,0),(0,1,0,1)}(\cdot)=2\alpha_1(\cdot)\beta_1(\cdot).$$

Thus $A_{11}=\mathbb{P}_{\alpha_1^2\mu_4+2\alpha_1\beta_1+\beta_1^2}$. Proceeding in the same way for the other matrices A_{ij} we find

$$\mathscr{A}_2=\begin{pmatrix} \mathbb{P}_{\alpha_1^2\mu_4+2\alpha_1\beta_1+\beta_1^2} & \mathbb{P}_{2\alpha_2(\alpha_1+\beta_1)\mu_4} & \mathbb{P}_{\alpha_2^2\mu_4} & \mathbb{P}_{2\omega(\alpha_1+\beta_1)\mu_4} & \mathbb{P}_{2\omega\alpha_2\mu_4} & \mathbb{P}_{\omega^2\mu_4} \\ \mathbb{P}_{\alpha_1+\frac{\beta_1}{\mu_4}} & \mathbb{P}_{\alpha_2} & 0 & \mathbb{P}_{\omega} & 0 & 0 \\ \mathbb{P} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbb{P}_{\alpha_1+\beta_1} & \mathbb{P}_{\alpha_2} & \mathbb{P}_{\omega} \\ 0 & 0 & 0 & \mathbb{P} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbb{P} \end{pmatrix}. \tag{36}$$

Similarly, when $p=2$, $q=1$ and $m=2$, we obtain

$$\mathscr{A}_2=\begin{pmatrix} \mathbb{P}_{\alpha_1^2\mu_4+2\alpha_1\beta_1+\beta_1^2} & \mathbb{P}_{2\beta_2(\alpha_1+\beta_1)\mu_4} & \mathbb{P}_{\beta_2^2\mu_4} & \mathbb{P}_{2\omega(\alpha_1+\beta_1)\mu_4} & \mathbb{P}_{2\omega\beta_2\mu_4} & \mathbb{P}_{\omega^2\mu_4} \\ \mathbb{P}_{\frac{\alpha_1+\beta_1}{\mu_4}} & \mathbb{P}_{\beta_2} & 0 & \mathbb{P}_{\omega} & 0 & 0 \\ \mathbb{P}_{\frac{1}{\mu_4}} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbb{P}_{\alpha_1+\beta_1} & \mathbb{P}_{\beta_2} & \mathbb{P}_{\omega} \\ 0 & 0 & 0 & \mathbb{P} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbb{P} \end{pmatrix}.$$

We are now in position to state the following result.

Theorem 4. *Let $p,q\geqslant 1$ and suppose $\rho\{\mathbb{P}_{C^{(m')}}\}<1$ for all $m'\leqslant 2m$, where $C^{(m')}(\cdot)$ is defined in Theorem* 1. *Let (ε_t) be the nonanticipative* 4*mth-order stationary solution of model* (1). *Then (ε_t^{2m}) is an* ARMA(N_2,N_2) *process, with*

$$N_2:=\binom{m+p+q-1}{p+q-1}\{d-n_0(\mathbb{P})\}-n_1(\mathscr{A}_m).$$

Remark 15. In the case $p = q = 1$ it can be seen, in view of Remark 12, that $\mathscr{A}_m = A_m$, where A_m is the $d(m+1) \times d(m+1)$ matrix of Section 4.1. Thus the results of Theorems 3 and 4 are the same in this special case.

Remark 16. Define f_{ij} such that $A_{ij} = \mathbb{P}_{f_{ij}}$ for $i,j = 1,\ldots,|\widetilde{\mathscr{R}}_m|$ and let $D_{ij} = \operatorname{diag}(f_{ij}(1),\ldots,f_{ij}(d))$. Then we have

$$\mathscr{A}_m = \begin{pmatrix} D_{11} & D_{12} & \cdots & D_{1|\widetilde{\mathscr{R}}_m|} \\ D_{21} & D_{22} & \cdots & D_{2|\widetilde{\mathscr{R}}_m|} \\ \vdots & \ddots & & \vdots \\ D_{|\widetilde{\mathscr{R}}_m|-1,1} & \cdots & & D_{|\widetilde{\mathscr{R}}_m|-1,|\widetilde{\mathscr{R}}_m|} \\ 0 & \cdots & 0 & I_d \end{pmatrix} \begin{pmatrix} \mathbb{P} & 0 & \cdots & 0 \\ 0 & \mathbb{P} & \cdots & 0 \\ \vdots & \ddots & & \vdots \\ 0 & \cdots & \mathbb{P} & 0 \\ 0 & \cdots & 0 & \mathbb{P} \end{pmatrix}. \tag{37}$$

It follows that

$$\det(\mathscr{A}_m) = \{\det(\mathbb{P})\}^{|\widetilde{\mathscr{R}}_m|} \prod_{k=1}^{d} \det f(k), \tag{38}$$

where $f(.)$ is the $(|\widetilde{\mathscr{R}}_m| - 1) \times (|\widetilde{\mathscr{R}}_m| - 1)$ matrix with (i,j)th element $f_{ij}(.)$, and that $n_0(\mathscr{A}_m) \geqslant |\widetilde{\mathscr{R}}_m| n_0(\mathbb{P})$, with equality if and only if $\det f(\cdot) \neq 0$.

Proof. Again, the proof is divided into three steps:

(i) We first derive the autocovariance function $\gamma_{\varepsilon^{2m}}(\ell)$ of ε^{2m}.
For $\ell > 1$, $p > 1$ and $q > 1$ we have, by Lemmas 3 and 4

$$\begin{aligned}
&\pi(k)E[\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}|\Delta_t = k] \\
&\quad = \mu_{2m_0} E[h_t^{m_0}\varepsilon_{t-1}^{2m_1}\ldots\varepsilon_{t-q+1}^{2m_{q-1}}h_{t-1}^{n_1}\ldots h_{t-p+1}^{n_{p-1}}\varepsilon_{t-\ell}^{2m}|\Delta_t = k]\pi(k) \\
&\quad = \mu_{2m_0}\sum_{\underline{l}\in\mathscr{S}_{m_0}} \frac{m_0!}{\Pi_{j=0}^{p+q} l_j!} E[\omega(\Delta_t)^{l_0}\alpha_1(\Delta_t)^{l_1}\ldots\alpha_q(\Delta_t)^{l_q}\beta_1(\Delta_t)^{l_{q+1}}\ldots\beta_p(\Delta_t)^{l_{p+q}} \\
&\qquad \times\varepsilon_{t-1}^{2(l_1+m_1)}\varepsilon_{t-2}^{2(l_2+m_2)}\ldots\varepsilon_{t-q+1}^{2(l_{q-1}+m_{q-1})}\varepsilon_{t-q}^{2l_q} \\
&\qquad \times h_{t-1}^{l_{q+1}+n_1}\ldots h_{t-p+1}^{l_{p+q-1}+n_{p-1}}h_{t-p}^{l_{p+q}}\varepsilon_{t-\ell}^{2m}|\Delta_t = k]\pi(k) \\
&\quad = \sum_{\underline{l}\in\mathscr{S}_{m_0}} \xi_{\underline{m}',\underline{l}}(k)\sum_{j=1}^{d} E[\varepsilon_{t-1}^{2(l_1+m_1)}\ldots\varepsilon_{t-q}^{2(l_q+m_q)}\varepsilon_{t-1}^{2(l_{q+1}+n_1)} \\
&\qquad \times h_{t-2}^{l_{q+2}+n_2}\ldots h_{t-p}^{l_{p+q}+n_p}\varepsilon_{t-\ell}^{2m}|\Delta_{t-1} = j]p(j,k)\pi(j)
\end{aligned} \tag{39}$$

with $m_q = n_p = 0$. It can be seen that, in view of Remark 14, (39) remains valid for $p \geqslant 1$ and $q \geqslant 1$. Hence (39) generalizes (30), obtained for $p = q = 1$. Note that

$$\varepsilon_{t-1}^{2(l_1+m_1)}\ldots\varepsilon_{t-q}^{2(l_q+m_q)}\varepsilon_{t-1}^{2(l_{q+1}+n_1)}h_{t-2}^{l_{q+2}+n_2}\ldots h_{t-p}^{l_{p+q}+n_p} = \Upsilon_{t-1}(\underline{m}' \star \underline{l})$$

and that for $\underline{l} \in \mathscr{S}_{m_0}$, $\underline{m}' \star \underline{l}$ belongs to $\mathscr{R}_{m''}$, with $m'' = m' - l_0 \leqslant m'$. Hence (39) can be written as

$$E[\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}|\Delta_t = k]\pi(k) = \sum_{\underline{l}\in\mathscr{S}_{m_0}} \xi_{\underline{m}',\underline{l}}(k) \sum_{j=1}^{d} E[\Upsilon_{t-1}(\underline{m}' \star \underline{l})\varepsilon_{t-\ell}^{2m}|\Delta_{t-1} = j]p(j,k)\pi(j)$$

or, in vector form, as

$$\Pi_{E[\Upsilon_t(\underline{m}')\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} = \sum_{\underline{l}\in\mathscr{S}_{m_0}} \mathbb{P}_{\xi_{\underline{m}',\underline{l}}} \Pi_{E[\Upsilon_t(\underline{m}' \star \underline{l})\varepsilon_{t-\ell+1}^{2m}|\Delta_t=\cdot]} \quad \text{for all } \ell > 1. \tag{40}$$

Note that this formula is analogous to (31), obtained for $p = q = 1$. In view of (40) we have, for $\ell > 1$

$$\mathscr{W}_m(\ell) := \begin{pmatrix} \Pi_{E[\Upsilon_t\{\tau^{-1}(|\widetilde{\mathscr{R}}_m|)\}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} \\ \Pi_{E[\Upsilon_t\{\tau^{-1}(|\widetilde{\mathscr{R}}_m|-1)\}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} \\ \vdots \\ \Pi_{E[\Upsilon_t\{\tau^{-1}(1)\}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} \end{pmatrix} = \mathscr{A}_m \mathscr{W}_m(\ell-1). \tag{41}$$

Now remark that

$$\Pi_{E[\Upsilon_t\{\tau^{-1}(|\widetilde{\mathscr{R}}_m|)\}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} = \Pi_{E[\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}, \quad \Pi_{E[\Upsilon_t\{\tau^{-1}(1)\}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]} = \Pi_{E[\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}.$$

Hence, $\gamma_{\varepsilon^{2m}}(\ell) = \upsilon' \mathscr{W}_m(\ell)$, with $\upsilon' = (\underline{1}_d', 0, \ldots, 0, -E(\varepsilon_t^{2m})\underline{1}_d')$.

(ii) By Lemma 2, with $n_0 = n_0(\mathscr{A}_m)$, $\ell_0 = 1$, $n = d|\widetilde{\mathscr{R}}_m|$ and $\mathscr{P} = \mathscr{P}_{\mathscr{A}_m}$, we find that (ε_t^{2m}) admits an ARMA representation, whose orders are $(d|\widetilde{\mathscr{R}}_m| - n_0(\mathscr{A}_m) - n_1(\mathscr{A}_m), d|\widetilde{\mathscr{R}}_m| - n_1(\mathscr{A}_m))$ after cancellation of the roots of modulus 1 (it can be shown as in Step (ii) of the Proof of Theorem 2 that $\varepsilon_t^{2m} - E(\varepsilon_t^{2m})$ is purely nondeterministic). When $n_0(\mathbb{P}) = 0$, we get the announced result (with an over-identification of the AR order, as given in the theorem, if $\det f(k) = 0$ for some integers k. See Remark 16).

(iii) Now suppose $n_0(\mathbb{P}) \neq 0$. With the notations of the Proof of Theorem 2, we introduce the block-diagonal $|\widetilde{\mathscr{R}}_m|\{d - n_0(\mathbb{P})\} \times d|\widetilde{\mathscr{R}}_m|$ matrix $\Lambda_{\mathscr{A}_m} = \operatorname{diag}(\Lambda_{\mathbb{P}}, \ldots, \Lambda_{\mathbb{P}})$, and the matrices $\mathscr{A}_m^*$ and $\tilde{\mathscr{A}}_m$ such that $\mathscr{A}_m = \mathscr{A}_m^* \Lambda_{\mathscr{A}_m}$, $\tilde{\mathscr{A}}_m = \Lambda_{\mathscr{A}_m} \mathscr{A}_m^*$. We have

$$\tilde{\mathscr{W}}_m(\ell) := \Lambda_{\mathscr{A}_m} \mathscr{W}_m(\ell) = \tilde{\mathscr{A}}_m \tilde{\mathscr{W}}_m(\ell-1), \quad \forall \ell > 1.$$

Thus $\upsilon' = \tilde{\upsilon}' \Lambda_{\mathscr{A}_m}$ with $\tilde{\upsilon}' = (\underline{1}_{d-n_0(\mathbb{P})}', 0, \ldots, 0, -E(\varepsilon_t^{2m})\underline{1}_{d-n_0(\mathbb{P})}')$. Hence $\gamma_{\varepsilon^{2m}}(\ell) = \tilde{\upsilon}' \tilde{\mathscr{W}}_m(\ell)$. The proof is completed as that of Theorem 2 by applying Lemma 2. □

Example 4.1 (*continued*). When $p = 1$, $q = 2$ and $m = 2$, the quantities introduced in the proof take the form

$$\Upsilon_t(\underline{m}') = \varepsilon_t^{2m_0} \varepsilon_{t-1}^{2m_1},$$

$$\underline{m}' = (m_0, m_1) \in \widetilde{\mathscr{R}}_2 = \{(2,0),(1,1),(0,2),(1,0),(0,1),(0,0)\},$$

$$\mathscr{W}_m(\ell) = (\Pi_{E[\varepsilon_t^4 \varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}, \Pi_{E[\varepsilon_t^2 \varepsilon_{t-1}^2 \varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}, \Pi_{E[\varepsilon_{t-1}^4 \varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}, \Pi_{E[\varepsilon_t^2 \varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}, \Pi_{E[\varepsilon_{t-1}^2 \varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]}, \Pi_{E[\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot]})'.$$

Moreover, we have by (38), $\det \mathscr{A}_2 = (\det \mathbb{P})^6 \prod_{k=1}^d \alpha_2^4(k)\mu_4$. Thus the process (ε_t^4) is solution of an ARMA$[6\{d - n_0(\mathbb{P})\} - n_1(\mathscr{A}_2), 6\{d - n_0(\mathbb{P})\} - n_1(\mathscr{A}_2)]$ equation. If $\alpha_2(k) = 0$ for some k, the AR order can be reduced.

To be more specific suppose that $d = 1$ and take $\alpha_1 = \beta_1 = 0$. Then the model under consideration is the standard ARCH(2) model of the form : $\varepsilon_t = \sqrt{\omega + \alpha_2 \varepsilon_{t-2}^2}\eta_t$. We have, by (36),

$$\mathscr{A}_2 = \begin{pmatrix} 0 & 0 & \alpha_2^2\mu_4 & 0 & 2\omega\alpha_2\mu_4 & \omega^2\mu_4 \\ 0 & \alpha_2 & 0 & \omega & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \alpha_2 & \omega \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

so that

$$\mathscr{P}_{\mathscr{A}_2}(x) = (x^2 - \alpha_2)(x - \alpha_2)(x^2 - \alpha_2^2\mu_4)(x - 1).$$

Applying Theorem 4 we therefore obtain an ARMA(5,5) model for (ε_t^4), with AR polynomial: $(1 - \alpha_2 L^2)(1 - \alpha_2 L)(1 - \alpha_2^2\mu_4 L^2)$. Direct calculations confirm these results and allow to reduce the ARMA orders. We have $\varepsilon_t^2 = \omega + \alpha_2\varepsilon_{t-2}^2 + h_t(\eta_t^2 - 1)$. Hence

$$(1 - \alpha_2 L^2)\varepsilon_t^2 = \omega + u_t, \tag{42}$$

where $u_t = h_t(\eta_t^2 - 1)$. Moreover, from $\varepsilon_t^4 = \{\omega^2 + \alpha_2^2\varepsilon_{t-2}^4 + 2\omega\alpha_2\varepsilon_{t-2}^2\}\mu_4 + h_t^2(\eta_t^4 - \mu_4)$ we get $(1 - \alpha_2^2\mu_4 L^2)\varepsilon_t^4 = \{\omega^2 + 2\omega\alpha_2\varepsilon_{t-2}^2\}\mu_4 + v_t$, where $v_t = h_t^2(\eta_t^4 - \mu_4)$, and thus by (42)

$$(1 - \alpha_2^2\mu_4 L^2)(1 - \alpha_2 L^2)\varepsilon_t^4 = \omega^2\mu_4(1 + \alpha_2) + (1 - \alpha_2 L^2)v_t + 2\omega\alpha_2\mu_4 u_{t-2}.$$

It is straightforward to show that $(1 - \alpha_2 L^2)v_t + 2\omega\alpha_2\mu_4 u_{t-2}$ is a MA(2) process. Finally, (ε_t^4) is the solution of an ARMA(4,2) equation. The MA part can be obtained numerically from the first autocovariances of (ε_t^4), for given values of α_2 and μ_4.

To conclude, it should be noted that Theorem 4 is of invaluable help for more complex models or higher powers than those of this example. In such cases the direct method for determining the ARMA orders and AR polynomial cannot be applied. In general, when $d > 2$, the ARMA representations involve high orders and deriving the ARMA coefficients cannot be done by hand. A method for computing the

autocovariances of ε_t^{2m}, and an algorithm for deriving the ARMA representations in the GARCH(1,1) case are proposed in Francq and Zakoïan [15].

Acknowledgements

The authors are particularly grateful for the constructive comments and suggestions for improvements made by the referees, an Associate Editor and the Editor of the journal.

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